AAE 440: Spacecraft Attitude Dynamics
3 credits

1 Course Information/Description

This course introduces students to the fundamentals of spacecraft attitude dynamics, including various attitude representations, spacecraft rigid body kinematics and dynamics, attitude equations of motion, and analysis of their stability with analytical and numerical approaches.

2 Learning Resources and Reference Text

Brightspace Learning Management System: Access the course videos, materials, and resources via Purdue's Brightspace learning management system. Begin with the Start Here tab, which describes how the course Brightspace is organized. We strongly suggest that you explore and become familiar not only with the site navigation but with content and resources available for this course. See the Student Services widget on the campus homepage for resources such as Technology Help, Academic Help, Campus Resources, and Protect Purdue.

Textbook: None required.

3 Learning Outcomes

By the end of the course, students will be able to

- acquire fundamental knowledge about rigid body kinematics and dynamics;
- formulate equations of motion to mathematically describe spacecraft attitude motion in space;
- analyze the attitude motion's stability and other characteristics based on the equations of motion;
- develop original computing scripts using a programming language to numerically simulate the attitude motion;
- present and communicate their technical approaches to given problems, their analysis/numerical results, and their interpretation in a written format

4 Assignments

The course has homework assignments and exams.

4.1 Homework assignments

The course has homework assignments (1 problem set approximately every week).

4.2 Exams

The course has a midterm exam and a final exam.

5 Evaluation

Course grades will be determined on the basis of problem sets (70%) and exams (30%).

All homework assignments must be submitted on Gradescope as instructed. No email submissions will be accepted. Late submissions after the deadline will be accepted but incur the following penalty:

- 0% score deduction within 0–1 hours past the deadline (grace period);
- 20% score deduction within 1–24 hours past the deadline;
- 40% score deduction within 24–48 hours past the deadline;
- 60% score deduction within 48–72 hours past the deadline; and
- 100% score deduction after 72 hours past the deadline.

The instructor reserves the right to define the final numeric ranges for letter grades. However, the numerical ranges will be based on, and will not be more stringent than, the standard scale (A: 90-100%, B: 80-89%, C: 70-79%, D: 60-69%, F: below 60%).